

PLATFORM SOFTWARE · OPTICAL TRACKING

SKADI Tracking Software (API and Viewer)

Make the in-house SKADI optical 3D tracker directly usable by surgical-navigation companies and research groups through its software layer.

EVIDENCE STATE**Ongoing****PERIOD****2023.02 - present****PORTFOLIO TIER****Platform Software****PRIMARY CAPABILITY****Medical Navigation & Visualization**

Develop and maintain the SKADI API and SDK, the Viewer that shows device state and tracking results, and a 3D Slicer custom-application template that lets researchers start immediately.

Problem and system boundary

01 / SDK LAYERS

SDK layers

Device connection, marker and frame definitions, real-time streaming, and error states are exposed through the API with language bindings.

PROBLEM

An optical tracker is not used as bare hardware: coordinate frames, marker definitions, real-time streaming, and error states must be exposed

Develop and maintain the SKADI API and SDK, the Viewer that shows device state and tracking results, and a 3D Slicer custom-application template that lets researchers start immediately.

PUBLIC BOUNDARY

Device specifications, accuracy figures, customer lists, and sales numbers are company-owned and not published.

Interfaces are agreed with optical and hardware designers, customer navigation developers, and research users.

EVIDENCE STATE

Ongoing / Demonstration clip: the SKADI Viewer shows a tracked instrument over a skull phantom in the 3D model and section views.

Owned decisions and implementation

MY ROLE

Own the design, implementation, and maintenance of the API, SDK, and Viewer; wrote the 3D Slicer custom-application template; support customer integrations and enquiries.

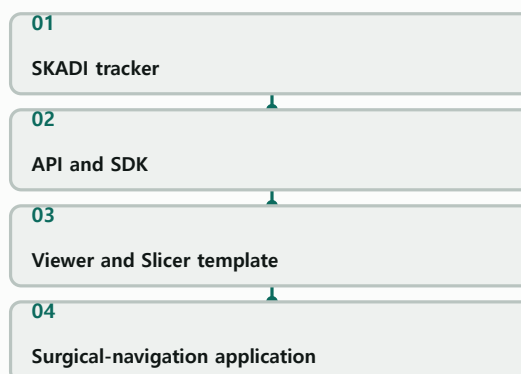
02 / VIEWER AND TEMPLATE

Viewer and template

The Viewer verifies device state and tracking output; the Slicer template lets researchers start a navigation

SYSTEM RELATIONSHIP

Software layers from tracker to application



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

SKADI / C++ / Python API / Viewer / 3D Slicer / Optical tracking

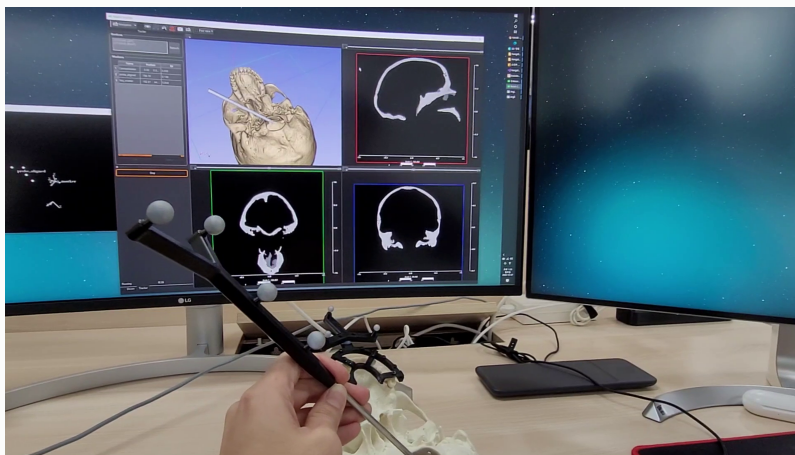
Verification and evidence ledger

03 / WORKING EVIDENCE

Working evidence

Working Viewer, API, and template screens are the evidence; customer-site footage is excluded.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

VIDEO / PUBLICLY INSPECTABLE

skadi-tracking-software-clip-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

skadi-tracking-software-poster-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

skadi-tracking-software-gallery-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

skadi-tracking-software-gallery-02

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

skadi-tracking-software-gallery-03

Approved derivative; caption in portfolio data.

Role, team result, and limits

04 / HARDWARE BOUNDARY

Hardware boundary

Specifications, accuracy, and sales are company-owned; this case covers the software layer only.

MY ROLE

Own the API, SDK, Viewer, and Slicer template.

Own the design, implementation, and maintenance of the API, SDK, and Viewer; wrote the 3D Slicer custom-application template; support customer integrations and enquiries.

TEAM RESULT

Device hardware, optical and mechanical design, sales, and delivery belong to other colleagues. Delivery records and revenue are company

Viewer screens, an API structure diagram, and the Slicer template in action are the evidence; customer-site footage is excluded.

LIMITATIONS

Device specifications, accuracy figures, customer lists, and sales numbers are company-owned and not published.

Demonstration clip: the SKADI Viewer shows a tracked instrument over a skull phantom in the 3D model and section views.

COLLABORATION BOUNDARY

Interfaces are agreed with optical and hardware designers, customer navigation developers, and research users.

Develop and maintain the SKADI API and SDK, the Viewer that shows device state and tracking results, and a 3D Slicer custom-application template that lets researchers start immediately.

Recap and joint-development invitation

Make the in-house SKADI optical 3D tracker directly usable by surgical-navigation companies and research groups through its software layer.

PRIMARY CAPABILITY

Medical Navigation & Visualization / 3D Geometry & Registration

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Interfaces are agreed with optical and hardware designers, customer navigation developers, and research users.

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